

Basics of Mechanical and Civil Engineering

Aaryan Khedekar

April 21, 2025

1 Thermodynamics

1.1 Energy:

1. Energy is constant and cannot be created or destroyed.
2. It can be divided into two types: Classical (Continuous) and Statistical (Discrete).
3. The amount of energy required to raise 1°F temperature of water at 68°F is called a **BTU** (British Thermal Unit).
4. The amount of energy required to raise 1°C temperature of water at 14.5°C is called a **Calorie**.
5. $1 \text{ cal} = 4.186 \text{ J}$, $1 \text{ BTU} = 1055 \text{ J}$.

1.1.1 System:

1. **System:** The thermodynamical region of interest.
2. **Surroundings:** The region outside the system.
3. **Types of System:**
 - (a) **Open System:** Mass and energy can be exchanged with the surroundings.
 - (b) **Closed System:** Energy can be exchanged with the surroundings.
 - (c) **Isolated System:** No exchange of mass or energy with the surroundings.
4. **Intensive Properties:** Properties that are independent of the mass of the system. (Examples: Temperature, Pressure, Density)
5. **Extensive Properties:** Properties that are dependent on the mass of the system. (Examples: Volume, Mass, Energy)

1.1.2 Zeroth Law of Thermodynamics:

If two systems are in thermal equilibrium with a third system, then they are in thermal equilibrium with each other.

1.1.3 Energies:

1. **Internal Energy (U):** The sum of all microscopic forms of energy in a system. It is a function of temperature.
2. **Kinetic Energy:** The energy due to motion. $KE = \frac{1}{2}mv^2$.
3. **Potential Energy:** The energy due to position. $PE = mgh$.

1.2 First Law of Thermodynamics:

1.2.1 Heat and Work:

1. **Heat:** The energy transfer by virtue of a temperature.
2. **Work:** The energy transfer by virtue of a force.
3. **Sign Conventions:**
 - (a) Heat given to system is positive; taken from the system is negative.
 - (b) Work done by the system is positive; done on the system is negative.
4. An isolated system is always in a state of equilibrium.
5. **Quasi-Static Equilibrium:** If two states are changed slowly enough, then it is assumed that system remains in equilibrium.
6. **The First Law:** $\delta Q = \Delta u + \delta W$.
7. **Question:** A quasi-static cyclic process can be represented as a circle of 1cm diameter on a PV diagram, where 1cm of P is equal to 300KPa and 1cm of V is equal to 0.1m³/kg. Find the work done.

Solution:

$$W = Area = \pi ab = \pi(150)(0.05) = 23.56kJ$$

8. **Question:** A classroom contains 55 students. Assume that each student emits 280 kJ/hr energy. The room is fixed with ten 100W bulbs. The heat entering through the walls and windows is approximately equal to 15000 kJ/hr. The AC used have 5kW cooling capacity. Find the number of AC units to be used if the temperature of the room is to be maintained at 21°C.

Solution:

$$Q_{in} = Q_{out} \implies (55 \times 280 \times \frac{1}{3600}) + (10 \times 100 \times 10^{-3}) + (15000 \times \frac{1}{3600}) = 5n$$
$$9.44 = 5n \implies n = 1.88 \approx 2$$

9. **Question:** A control system of mass 2.5 kg undergoes a quasi-static process that can be represented on a PV graph as $p = a + bv$. The initial and final pressures are 1000kPa and 200kPa respectively. The initial and final volumes are 0.2m³ and 1.2m³ respectively. The internal energy is given as $u = 1.5pv - 85kJ/kg$. Find the net heat transfer of the process.

Solution: Initially, $1000 = a + b(0.2)$

Finally, $200 = a + b(1.2)$.

Hence, $b = -800$, $a = 1160$.

Now,

$$W = \int_{0.2}^{1.2} p dv = \int_{0.2}^{1.2} (1160 - 800v) dv = 600kJ$$

Also,

$$u = 1.5(200)(1.2) - 1.5(1000)(0.2) = 60$$

$$\implies U = um = 60 \times 2.5 = 150kJ$$

Hence,

$$Q = U + W = 150 + 600 \implies Q = 750kJ$$

1.2.2 Steady Flow Energy Equation:

Consider a control volume system with an inlet at a height z_1 from the datum, letting in mass m_1 with velocity V_1 , specific volume v_1 and specific pressure p_1 . The outlet is at a height z_2 from the datum, letting out mass m_2 with velocity V_2 , specific volume v_2 and specific pressure p_2 . The heat transfer is Q and the work done is W . The internal energy is E .

Continuity Equation:

$$m_1 = m_2 \implies \rho_1 A_1 V_1 = \rho_2 A_2 V_2$$

Steady Flow Energy Equation:

$$\delta W_{net} = \delta W - p_1 v_1 dm_1 + p_2 v_2 dm_2$$

Considering control mass at this instant, we know $\delta Q = \Delta E + \delta W$

$$\delta Q = (e_2 - e_1)m + \delta W - p_1 v_1 dm_1 + p_2 v_2 dm_2$$

If m changes,

$$\delta Q = e_2 dm_2 - e_1 dm_1 + \delta W - p_1 v_1 dm_1 + p_2 v_2 dm_2$$

$$e_1 dm_1 + p_1 v_1 dm_1 + \delta Q = e_2 dm_2 + p_2 v_2 dm_2 + \delta W$$

Now, we know $\delta m_1 = \delta m_2 = \delta m$

$$e_1 + p_1 v_1 + \frac{\delta Q}{\delta m} = e_2 + p_2 v_2 + \frac{\delta W}{\delta m}$$

Now,

$$e_1 = u_1 + \frac{V_1^2}{2} + gz_1, \quad e_2 = u_2 + \frac{V_2^2}{2} + gz_2$$

$$u_1 + \frac{V_1^2}{2} + gz_1 + p_1 v_1 + \frac{\delta Q}{\delta m} = u_2 + \frac{V_2^2}{2} + gz_2 + p_2 v_2 + \frac{\delta W}{\delta m}$$

This is the steady flow energy equation.

1.2.3 Enthalpy:

Enthalpy (H): $h = u + pv$, where h is the specific enthalpy.

$$h_1 + \frac{V_1^2}{2} + gz_1 + \frac{\delta Q}{\delta m} = h_2 + \frac{V_2^2}{2} + gz_2 + \frac{\delta W}{\delta m}$$

If $V_1 = 0, V_2 = 0, z_1 = 0, z_2 = 0$, then

$$h_1 + \frac{\delta Q}{\delta m} = h_2 + \frac{\delta W}{\delta m}$$

For $\frac{dQ}{dm} = 0$,

$$W = m(h_1 - h_2)$$

1.2.4 Nozzle:

If $z_1 = z_2, \delta Q = 0, \delta W = 0$, then

$$h_1 + \frac{V_1^2}{2} = h_2 + \frac{V_2^2}{2}$$

1.3 Second Law of Thermodynamics:

1.3.1 Definitions:

1. **Kelvin-Planck's Statement:** It is impossible to construct a device with only one heat reservoir that produces work.
2. **Clausius Statement:** It is impossible to construct a device that transfers heat from a cold body to a hot body without any work.
3. **Carnot's Statement:** No engine can be more efficient than a reversible engine working between the same temperature limits.
4. PMM2 is impossible.

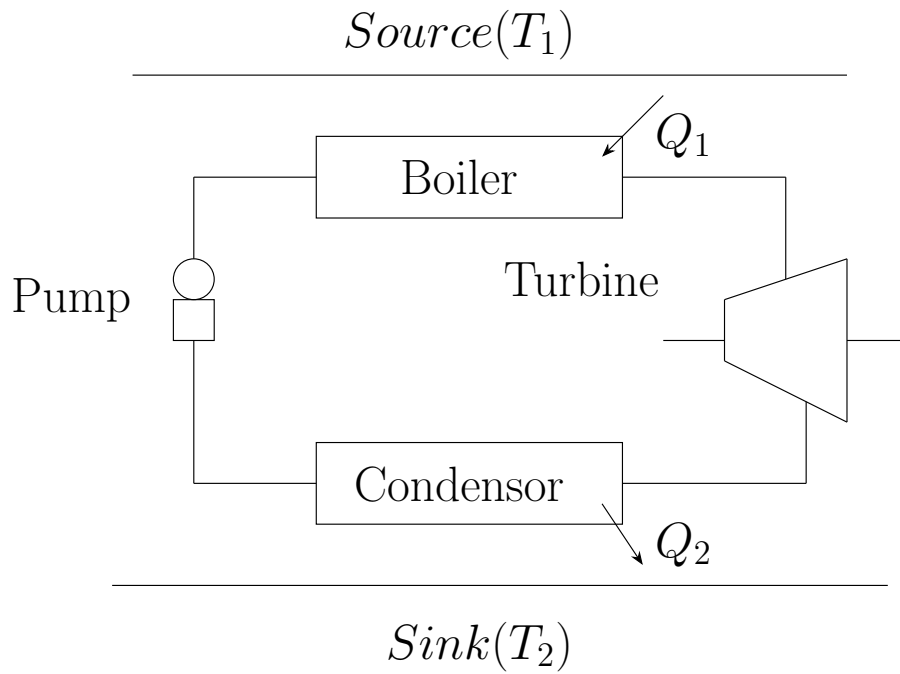
1.3.2 Heat Engine:

Efficiency of heat engine:

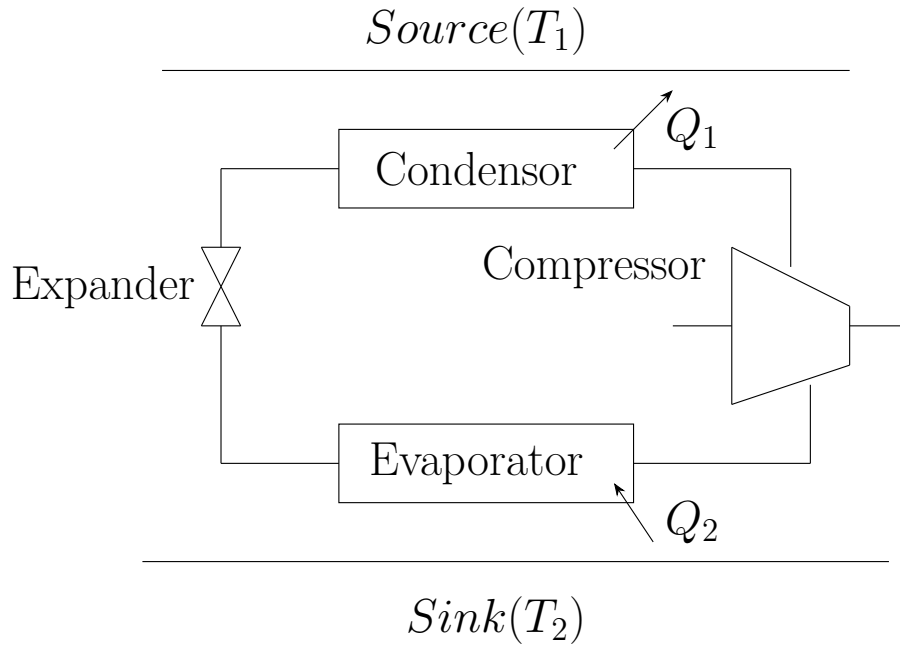
$$\eta = \frac{W_{net}}{Q_1} = \frac{Q_1 - Q_2}{Q_1} = 1 - \frac{Q_2}{Q_1}$$

Coefficient of Performance of heat engine:

$$\alpha = \frac{Q_1}{W_{net}} = \frac{Q_1}{Q_1 - Q_2} = \frac{1}{\eta}$$



1.3.3 Refrigerator:



Coefficient of Performance of refrigerator:

$$\beta = \frac{Q_2}{W_{net}} = \frac{Q_2}{Q_1 - Q_2}$$

$$\alpha = \frac{Q_1}{Q_1 - Q_2} = \frac{Q_1 - Q_2 + Q_2}{Q_1 - Q_2} = 1 + \frac{Q_2}{Q_1 - Q_2}$$

Hence,

$$\alpha = 1 + \beta$$

1.3.4 Carnot's Cycle:

For a reversible engine,

$$\eta_{rev} = 1 - \frac{Q_2}{Q_1} = 1 - \frac{T_2}{T_1}$$

where T is the absolute temperature.

1. **Question:** An engineer claims that he built an engine whose input is 105MJ at 400K, and it rejects heat of 42MJ at 200K. It derives a power of 12kWh. Would you invest in the engine?

Solution:

$$\eta = 1 - \frac{T_2}{T_1} = 1 - \frac{200}{400} = 0.5$$

$$W_{net} = \eta \times Q_1 = (0.5)(105) = 52.5 MJ$$

$$\therefore P_{net} = 52.5 \times \frac{5}{18} kWh = 14.583 kWh$$

Now, an the maximum power derived by any engine working between the given temperature limits is 14.583kWh. Hence, the engine is not worth investing in.

2. **Question:** For a reversible engine, if we want to increase the efficiency, then which is better: Increasing Source T_1 or Decreasing Sink T_2 ?

Solution: We know that $\eta = 1 - \frac{T_2}{T_1}$.

If we increase T_1 , then considering T_2 as constant,

$$d\eta_1 = T_2 \times d\left(\frac{1}{T_1}\right) = \frac{T_2}{T_1^2} dT$$

If we decrease T_2 , then considering T_1 as constant,

$$d\eta_2 = \frac{1}{T_1} d(T_2) = \frac{1}{T_1} dT$$

$$d\eta_1 = \frac{T_2}{T_1} d\eta_2 \implies \eta_2 > \eta_1$$

Hence, decreasing T_2 is better.

3. **Question:** A carnot engine is working between a source of T_1 K and a sink of T_2 K. Another carnot engine is working between T_1 - T_3 and T_3 - T_2 . Compare the efficiencies.

Solution For engine 1,

$$\eta_1 = 1 - \frac{T_2}{T_1}$$

For engine 2, consider W_2 work is done between T_1 - T_3 (Q_1 , Q_2) and W_3 work is done between T_3 - T_2 (Q_2 , Q_3).

$$\eta_{2a} = 1 - \frac{T_3}{T_1}, \eta_{2b} = 1 - \frac{T_2}{T_3}$$

Also,

$$\eta_2 = \frac{W_{net}}{Q_1} = \frac{W_2 + W_3}{Q_1} = \eta_{2a} + \frac{W_3}{Q_1}$$

Now,

$$\frac{W_3}{Q_1} = \frac{Q_2 - Q_3}{Q_1} = \frac{T_3}{T_1} - \frac{T_3 \cdot T_2}{T_1 \cdot T_3} = \frac{T_3 - T_2}{T_1} = \eta_{2b} \times \frac{T_3}{T_1} = \eta_{2b}(1 - \eta_{2a})$$

Hence,

$$\eta_2 = \eta_{2a} + \eta_{2b}(1 - \eta_{2a})$$

$$\eta_2 = 1 - \frac{T_3}{T_1} + 1 - \frac{T_2}{T_3} - \left(1 - \frac{T_3}{T_1}\right) \left(1 - \frac{T_2}{T_3}\right) = 1 - \frac{T_2}{T_1}$$

Hence, the efficiencies are equal.

4. **Question:** A carnot engine is built such that W_1 work is done between reserve T_1 and T . W_2 work is done between reserves T and T_2 . If the efficiency of both cycles are same, find T in terms of T_1 and T_2 . If the work done by both engines is same, find T in terms of T_1 and T_2 .

Solution: For the first part,

$$\eta_1 = 1 - \frac{T}{T_1}, \eta_2 = 1 - \frac{T_2}{T}$$

$$\eta_1 = \eta_2 \implies 1 - \frac{T}{T_1} = 1 - \frac{T_2}{T} \implies T = \sqrt{T_1 T_2}$$

Hence, T is the geometric mean of T_1 and T_2 .

For the second part,

$$W_1 = Q_1 - Q_2 = T_1 - T$$

$$W_2 = Q_2 - Q_3 = T - T_2$$

$$W_1 = W_2 \implies T_1 - T = T - T_2 \implies T = \frac{T_1 + T_2}{2}$$

Hence, T is the arithmetic mean of T_1 and T_2 .

1.4 Entropy:

1. For a reversible process,

$$\frac{Q_1}{T_1} = \frac{Q_2}{T_2} \implies \sum_{cycle} \frac{Q}{T} = 0 \implies \oint \frac{dQ}{T} = 0$$

2. The term $\oint \frac{dQ}{T}$ is called as "entropy", denoted by "S".
 3. For an irreversible process, $Q_2 > Q_1 \implies \oint \frac{dQ}{T} < 0$.
 4. This is called as the "Classius Inequality":

$$\boxed{\oint \frac{dQ}{T} \leq 0}$$

5. **Question:** In a reversible process, a heat of 750kJ is added to a liquid-vapour mixture at a temperature of 300K and constant pressure. This process converts a part of the liquid into vapour. Find the entropy change of the process.

Solution:

$$\Delta S = \frac{\Delta Q}{\Delta T} = \frac{750}{300} = 2.5 kJ/K$$

6. Entropy is the measure of disorderness in the system.

$$S_1 - S_2 = \int \frac{dQ}{T} + S_{gen}$$

where S_{gen} is the entropy generation (≥ 0)

7. **According to Third Law of Thermodynamics**, For a pure crystal, when $T = 0$, $S = 0$.

8. For an isolated system,

$$\Delta S_{universe} = \Delta S_{system} + \Delta S_{surroundings} \geq 0$$

2 Fluid Mechanics:

Pressure, Temperature, Volume, Density, Viscosity, Surface Tension, etc. are Properties of Fluids.

2.1 Viscosity:

1. Velocity Gradient:

$$\frac{du}{dy} = \frac{v}{l}$$

2. Shear Strain:

$$\theta = \frac{vt}{l}$$

3. Deformation Rate:

$$\frac{d\theta}{dt} = \frac{du}{dy} = \frac{v}{l}$$

4. If a Force F is acting along the plane of area A, then Shear Stress:

$$\tau = \frac{F}{A}$$

$$\tau \propto \frac{d\theta}{dt} \implies \tau \propto \frac{du}{dy} \implies \tau = \mu \frac{du}{dy}$$

where μ is the Coefficient of Viscosity.

$$\frac{F}{A} = \mu \frac{du}{dy} \implies F = \frac{\mu Av}{t}$$

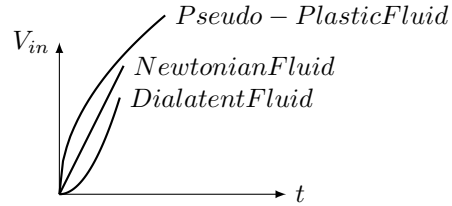
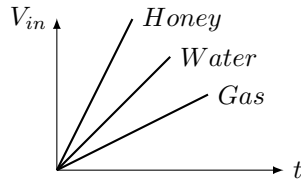
5. For a cylinder of radius R, thickness t, length L, moving with velocity v, angular velocity ω , having a viscosity coefficient μ , then Torque:

$$T = F \times R = \left(\frac{\mu Av}{t} \right) R = \frac{\mu(2\pi RL)(R\omega)R}{t}$$

$$T = \frac{2\pi\mu R^3\omega L}{t} \implies \boxed{\mu = \frac{Tt}{2\pi R^3\omega L}}$$

6. For Newtonian Fluids,

$$\tau \propto \frac{du}{dy}$$



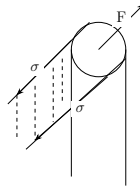
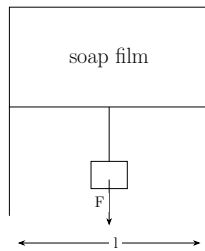
7. **Question:** A block is placed on a wedge with an oil film of thickness t and viscosity μ . The weight of the body is W . The surface area of the bottom of the block is A . Find its terminal velocity.

Solution: From the FBD,

$$W \sin \theta = \frac{\mu A v}{t} \implies v = \frac{W t \sin \theta}{\mu A}$$

2.2 Surface Tension:

1. Below is a simple surface tension experiment



$$F = \sigma(2l) \implies \sigma = \frac{F}{2l}$$

2. Units of σ is $\text{N/m} = \text{Nm/m}^2 = \text{U/A}$
Hence, surface tension is referred as surface energy per unit area.
3. For a sphere of radius r , $\sigma = \frac{F}{2\pi r}$
4. For a thin film of length l , $\sigma = \frac{F}{2l}$
5. For a thin film,

$$\sigma(2\pi r) = F = \Delta p(\pi r^2) \implies \Delta p = \frac{2\sigma}{r}$$

6. For a liquid bubble,

$$\Delta p = \frac{4\sigma}{r}$$

7. For a surface with two radii of curvatures r_1 and r_2 ,

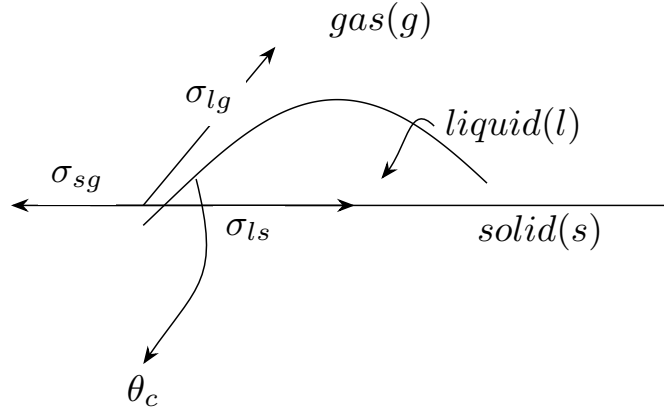
$$\Delta p = \sigma \left(\frac{1}{r_1} + \frac{1}{r_2} \right)$$

This is called as "Young-Laplace Equation".

8. Surface Tension of Water = 0.073 N/m

2.2.1 Contact Angle:

1. Surface Tension is not a singular property. It also depends upon the other surface of contact.



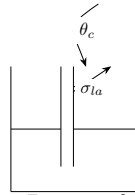
2. θ_c is the "contact angle".

3. Young's Equation:

$$\sigma_{lg} = \sigma_{ls} + \sigma_{sg} \cos \theta_c$$

4. Teflon Coating has a contact angle of $\approx 180^\circ$.
5. Silver Coating has a contact angle of $\approx 0^\circ$.

2.2.2 Capillary Effect:



$$(\sigma_{la}(2\pi r)\cos\theta = W_l = (\pi r^2 h)\rho g)$$

$$h = \frac{2\sigma_{la}\cos\theta}{\rho g r}$$

Question: Roots of a tree are of diameter 0.005mm, contact angle of water is 15° and surface tension is 0.073N/m at 20° C. How high the roots can hold the water?

Solution:

$$h = \frac{2\sigma \cos \theta}{\rho g r} = \frac{2 \times 0.073 \times \cos(15^\circ)}{10^3 \times 10 \times 2.5 \times 10^{-6}} = 5.64m$$

2.2.3 Surface Tension of Bubble Inside Surface:

$$(\sigma_{sa} \sin \theta)(2\pi R \sin \theta) = W$$

$$W = 2\pi r \sigma_{sa} \sin^2 \theta$$

2.3 Pressure:

1. For a uniform fluid at rest, $\Delta p = p_1 - p_0 = \rho gh$.
- 2.

$$\vec{V} = \frac{\partial x}{\partial t} \hat{i} + \frac{\partial y}{\partial t} \hat{j} + \frac{\partial z}{\partial t} \hat{k} = u \hat{i} + v \hat{j} + w \hat{k}$$

$$\vec{a} = \frac{dV}{dt} = \frac{\partial u}{\partial x} \frac{\partial x}{\partial t} \hat{i} + \frac{\partial v}{\partial y} \frac{\partial y}{\partial t} \hat{j} + \frac{\partial w}{\partial z} \frac{\partial z}{\partial t} \hat{k} + \frac{\partial \vec{V}}{\partial t}$$

$$\therefore \vec{a} = (\vec{\nabla} \cdot \vec{V}) \vec{V} + \frac{\partial \vec{V}}{\partial t}$$

2.4 Navier-Stoke's Equation:

$$\frac{1}{\rho} \frac{D\vec{v}}{Dt} = -\vec{\nabla} p + \mu \nabla^2 \vec{v} + g \vec{z}$$

3 Stress Analysis:

3.1 Some Terminologies:

1. **Hooke's Law:**

$$\sigma \propto \epsilon \implies \sigma = E\epsilon \text{ (E is young's modulus.)}$$

$$\tau \propto \gamma \implies \tau = G\gamma \text{ (G is shear's modulus.)}$$

2. **Poisson's Effect:** If a material is stretched in one direction, it contracts in the other direction to maintain constant volume.

$$\epsilon_y = -\nu\epsilon_x, \epsilon_z = -\nu\epsilon_x$$

3. **Small Strain Phenomenon:** For strains much smaller than characteristic dimensions:

$$\epsilon_v = \frac{\Delta V}{V} = \frac{\Delta l}{l} + \frac{\Delta b}{b} + \frac{\Delta h}{h} \implies \epsilon_v = \epsilon_x + \epsilon_y + \epsilon_z$$

4. **Bulk Modulus:**

$$\kappa = \frac{\sigma}{\Delta V/V} = \frac{\sigma}{\epsilon_v} \implies \sigma = \kappa\epsilon_v$$

$$\kappa = -V \left(\frac{\Delta P}{\Delta V} \right)_T = -V \frac{\partial P}{\partial V}$$

5. **Velocity of Sound:**

$$c = \sqrt{\frac{\kappa}{\rho}}$$

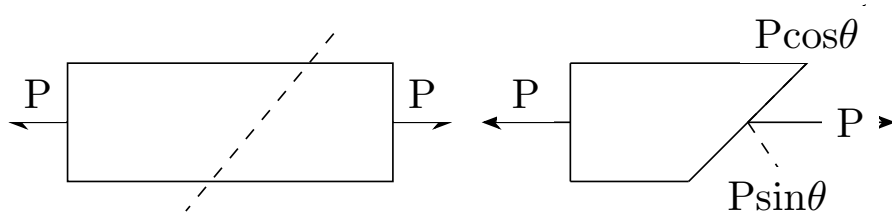
6. We know if v is specific volume, then $\rho v = 1$

$$\therefore \frac{d\rho}{\rho} + \frac{dv}{v} = 0 \implies \kappa = \rho \frac{\partial p}{\partial \rho}$$

κ is called as the "coefficient of compressibility".

If $\kappa \rightarrow \infty$, then the material is incompressible.

7. **Cutting at an angle:**



$$\sigma = \frac{P \sin \theta}{A / \sin \theta} = \frac{P}{A} \sin^2 \theta$$

$$\tau = \frac{P \cos \theta}{A / \sin \theta} = \frac{P}{A} \sin \theta \cos \theta = \frac{P}{2A} \sin(2\theta)$$

8. **Steel:** $E = 210 \text{ GPa}$, $\sigma_y = 340 \text{ MPa}$.
Aluminium: $E = 70 \text{ GPa}$, $\sigma_y = 120 \text{ MPa}$.

9.

$$\epsilon_{poisson} = \mu \frac{\Delta l}{l} \implies \epsilon_{poisson} = \mu \epsilon$$

10.

$$\epsilon_x = \frac{\epsilon_x}{E} - \mu \frac{\epsilon_y}{E} - \mu \frac{\epsilon_z}{E}, \tau_{xy} = \frac{1}{G} \gamma_{xy}$$

The above are called as the "Generalized Hooke's Law".

11. **Axial Stiffness:**

$$\epsilon = \frac{\Delta l}{l} = \frac{\sigma}{E} \implies \frac{F}{\Delta l} = \frac{EA}{l}$$

$\frac{F}{\Delta l}$ is called as the "axial stiffness" (k_a).

12. **2-D Stress:** Consider a thin cylinder. If pressure is P , thickness is t , axial stress is σ_a , radius is r , and hooke's stress is σ_h , then:

$$F_a = P(\pi r^2) = \sigma_a(2\pi r)t \implies \sigma_a = \frac{Pr}{2t}$$

$$F_h = P(2r)l = 2 \times \sigma_h(lh) \implies \sigma_h = \frac{Pr}{t}$$

13.

$$E = 2G(1 + \nu)$$

14.

$$E = 2\kappa(1 - 2\nu)$$

15. **Engineering Stress & Strain:**

$$\sigma = \frac{F}{A_0}, \quad \epsilon = \frac{\Delta l}{l_0}$$

where A_0 and l_0 are initial area of cross section and length.

16. **True Strain:**

$$d\epsilon = \frac{dl}{l} \implies \epsilon = \ln(l) \Big|_{l_0}^l \implies \epsilon_{true} = \ln\left(\frac{l}{l_0}\right)$$

$$\therefore \epsilon_{true} = \ln\left(\frac{l_0 + \Delta l}{l_0}\right) = \ln\left(1 + \frac{\Delta l}{l_0}\right) \implies \epsilon_{true} = \ln(1 + \epsilon_{engg})$$

17. **True Stress:** Assuming no volume change, $A_0 l_0 = A l \implies \frac{A_0}{A} = \frac{l}{l_0}$

$$\sigma = \frac{F}{A} = \frac{F}{A_0} \cdot \frac{A_0}{A} = \frac{F}{A_0} \cdot \frac{l_0}{l} \implies \boxed{\sigma_{true} = \sigma_{engg}(1 + \varepsilon_{engg})}$$

18. **Thermal Strain:**

(a) For no constraint, $\varepsilon = \alpha \Delta T$ and $\sigma = 0$.

This is called as "free expansion".

(b) For constraint, $\varepsilon = 0$ and $\sigma = E \alpha \Delta T$

19. **Torsion:** For a cylinder of radius R, length l, applied torque T, producing shear strain γ at angle θ

$$R\theta = l\gamma \implies \gamma_{max} = \frac{R\theta}{l} \implies \tau_{max} = \frac{GR\theta}{l}$$

Now,

$$\frac{\gamma_{max}}{\gamma} = \frac{R}{r} \implies \gamma = \gamma_{max} \frac{r}{R} \implies \tau = \frac{Gr\gamma_{max}}{R}$$

20. **Rotation Matrix:**

$$R = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

For a vector, $R\vec{r} = \vec{r}'$

For a tensor σ , $R^T \sigma R = \sigma'$

$$\therefore \begin{bmatrix} \sigma_{x'} & \tau_{x'y'} \\ \tau_{x'y'} & \sigma_{y'} \end{bmatrix} = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} \sigma_x & \tau_{xy} \\ \tau_{xy} & \sigma_y \end{bmatrix} \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

On solving the matrices,

$$\sigma_{x'} = \frac{\sigma_x + \sigma_y}{2} + \left(\frac{\sigma_x - \sigma_y}{2} \right) \cos(2\theta) + \tau_{xy} \sin(2\theta)$$

$$\sigma_{y'} = \frac{\sigma_x + \sigma_y}{2} - \left(\frac{\sigma_x - \sigma_y}{2} \right) \cos(2\theta) - \tau_{xy} \sin(2\theta)$$

$$\tau_{x'y'} = - \left(\frac{\sigma_x - \sigma_y}{2} \right) \sin(2\theta) + \tau_{xy} \cos(2\theta)$$

For maximum / principal stresses,

$$\sigma_1, \sigma_2 = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2} \right)^2 + \tau_{xy}^2}$$

If $\sigma_1 > \sigma_y$, failure occurs.

4 Heat Transfer:

4.1 Conduction:

1. It is the heat transferred due to physical contact of two bodies.
2. Consider a wall with boundary temperatures T_1 and T_2 and a small width dx across with an area dA . Let $\dot{Q}$ be the rate of heat transfer. Then according to Fourier,

$$\dot{Q} \propto A, \quad \dot{Q} \propto \frac{1}{\Delta x}, \quad \dot{Q} \propto \Delta T = (T_1 - T_2)$$

Hence,

$$\dot{Q} \propto A \frac{\Delta T}{\Delta x} \implies \boxed{\dot{Q} = -kA \frac{\Delta T}{\Delta x} = -kA \frac{dT}{dx}}$$

This is called as "Fourier Law of conduction" where k is called as "thermal conductivity".

3.

$$\begin{aligned} -kA \left(\frac{\partial^2 T}{\partial x^2} \right) - kA \left(\frac{\partial^2 T}{\partial y^2} \right) - kA \left(\frac{\partial^2 T}{\partial z^2} \right) &= A\rho C_p \frac{\Delta T}{\Delta t} \\ \implies \frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} &= -\frac{\rho C_p}{k} \frac{\partial T}{\partial t} \\ \text{Let } \frac{\rho C_p}{k} = \alpha &\implies \boxed{\nabla^2 T = -\frac{1}{\alpha} \frac{\partial T}{\partial t}} \end{aligned}$$

where α is called as "diffusivity".

4. **Question:** For a wall with $k = B + Cx$, find the value of ΔT at a distance x in the wall.

Solution:

$$\dot{Q} = -kA \frac{\Delta T}{\Delta x} = -(B + Cx)A \frac{dT}{dx} \implies dT = -\frac{\dot{Q}}{A(B + Cx)} dx$$

$$\text{Integrating, } \Delta T = \int dT = -\frac{\dot{Q}}{A} \int_0^x \frac{1}{(B + Cx)} dx$$

$$\Delta T = -\frac{\dot{Q}}{AC} \ln(B + Cx) \Big|_0^x = \Delta T = -\frac{\dot{Q}}{AC} \ln \left| 1 + \frac{Cx}{B} \right|$$

4.2 Convection:

1. There are two types of convection:
 - (a) Natural Convection
 - (b) Forced Convection

2.

$$\dot{Q} \propto (T_s - T_\infty), \quad \dot{Q} \propto A$$

$$\therefore \dot{Q} \propto A(T_s - T_\infty) \implies \boxed{\dot{Q} = hA(T_s - T_\infty)}$$

This is called the "Newton's law of cooling" and h is called as the "convection coefficient".

4.2.1 Lumped Mass Model:

Assume whole body to be at the same temperature T_b and the ambient outside temperature be T_∞ . Let A be the surface area of the body.

$$\dot{Q} = hA(T_s - T_\infty) = -\rho C_p \frac{dT}{dt}$$

$$\int_{T_b}^T \frac{dT}{T - T_\infty} = - \int_0^t \frac{hA}{\rho C_p} dt \implies \boxed{\ln \left| \frac{T - T_\infty}{T_b - T_\infty} \right| = - \frac{hA}{\rho C_p} t}$$

4.3 Radiation:

$$\dot{Q} = \sigma \varepsilon A T^4$$

where $0 \leq \varepsilon \leq 1$ and for a black body, $\varepsilon = 1$

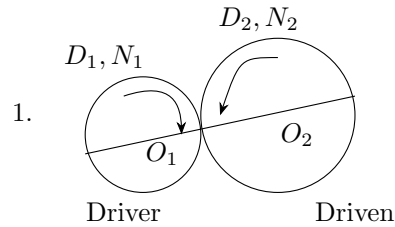
5 Power Transmission

5.1 Gears:

5.1.1 Types of Gears:

1. **Spur Gears:** These transfer energy between two parallel shafts.
2. **Helical Gears:** These are similar to spur gears, however they are in a helical shape.
3. **Revel Gears:** These transfer energy between two nonparallel shafts. These are used for intersecting shafts.
4. **Rack and Pinion:** In a gear train system, the largest gear is called as the "rack"; whereas the smallest gear is called as the "pinion".

5.1.2 Law of Gearing:



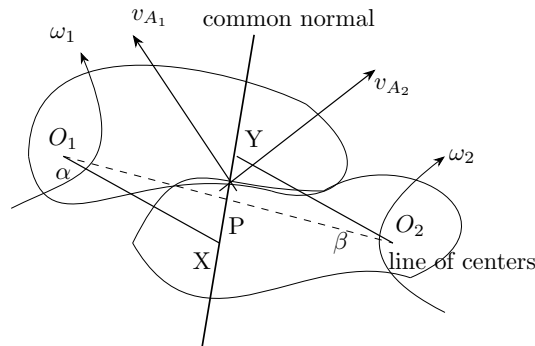
If N is the angular velocity in rps
($N = 2\pi\omega$).

For pure rolling v_A is at same velocity.

$$(v_A)_1 = (v_A)_2 \implies (r\omega)_1 = (r\omega)_2$$

$$\frac{D_1}{2} \frac{N_1}{2\pi} = \frac{D_2}{2} \frac{N_2}{2\pi} \implies \boxed{N_1 D_1 = N_2 D_2}$$

2. If the radius of gear is r , the limited friction is μ_l and pressure is P , then the maximum amount of energy transferred is $\boxed{E_{max} = \mu_l r P}$
3. Consider the following gear system:



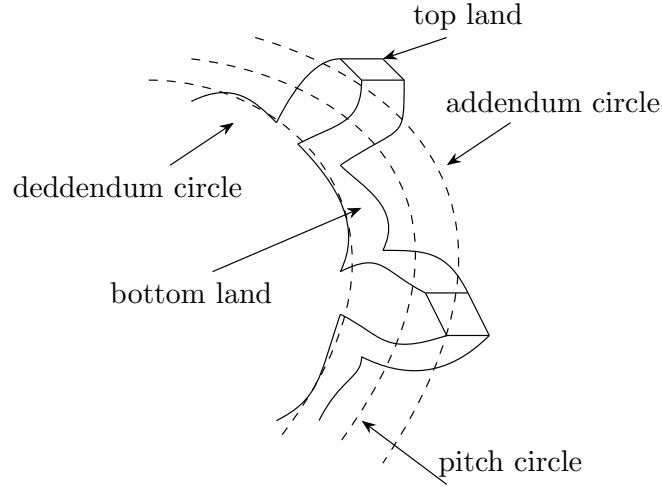
$$v_{A_1} \cos \alpha = v_{A_2} \cos \beta \implies (O_1 A) \omega_1 \cos \alpha = (O_2 A) \omega_2 \cos \beta$$

$$\therefore (O_1 A \cos \alpha) \omega_1 = (O_2 A \cos \beta) \omega_2 \implies (O_1 X) \omega_1 = (O_2 Y) \omega_2$$

Now, $\triangle XO_1P$ and $\triangle YO_2P$ are similar triangles, hence:

$$(O_1 P) \omega_1 = (O_2 P) \omega_2 \implies \boxed{\frac{\omega_1}{\omega_2} = \frac{O_2 P}{O_1 P}}$$

4. **The Law of Gearing:** The Pitch Point "P" (Point of intersection of common normal and line of centres) is fixed.
5. Involute and cycloids are two curves that satisfy the gearing law.
6. Consider the below figure:

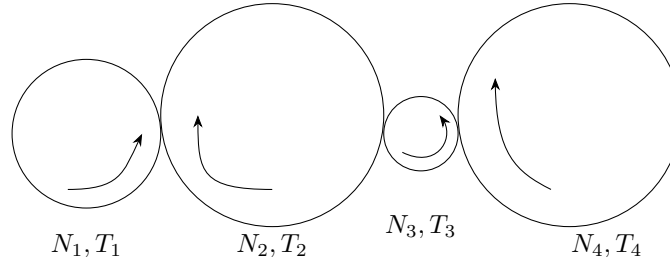


- (a) The locus of the pitch points is called the "pitch circle".
- (b) The area below the pitch circle is called "bottom land" and the locus of most bottom land is called as the "dedendum circle".
- (c) The area above the pitch circle is called "top land" and the locus of most top land is called as the "addendum circle".
- (d) If D is the diameter of pitch circle, and T is number of teeth of the gear, then circular pitch: $p = \frac{\pi D}{T}$.
- (e) Module: $m = \frac{D}{T}$. For a gear system, module is constant.
- (f) Diameter Pitch: $p_d = \frac{T}{D}$.
- (g)

$$m = \frac{D_1}{N_1} = \frac{D_2}{N_2} \text{ and } N_1 D_1 = N_2 D_2 \implies \boxed{\frac{D_1}{D_2} = \frac{T_1}{T_2} = \frac{N_2}{N_1}}$$

5.1.3 Gear Trains:

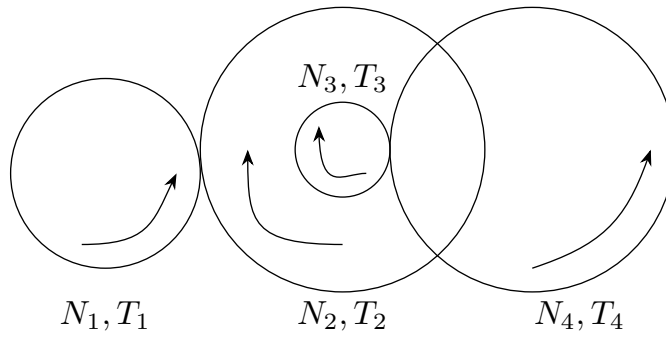
1. Simple Gear Trains:



$$\frac{N_1}{N_2} = \frac{T_2}{T_1}, \quad \frac{N_2}{N_3} = \frac{T_3}{T_2}, \quad \frac{N_3}{N_4} = \frac{T_4}{T_3} \Rightarrow \boxed{\frac{N_1}{N_4} = \frac{T_4}{T_1}}$$

Here the middle gears are called as "idler gears".

2. Compound Gear Trains:



$$\frac{N_1}{N_2} = \frac{T_2}{T_1}, \quad N_2 = N_3, \quad \frac{N_3}{N_4} = \frac{T_4}{T_2} \Rightarrow \boxed{\frac{N_1}{N_4} = \frac{T_2 T_4}{T_1 T_3}}$$

3.

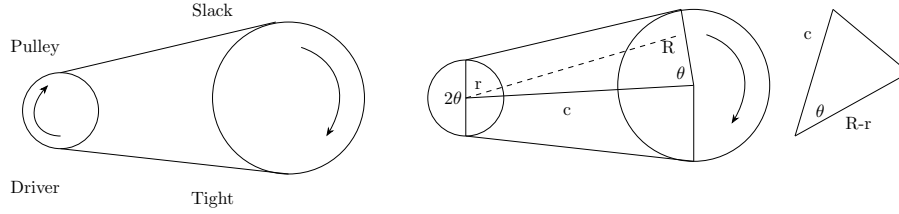
$$\frac{D_1 + D_2}{2} = \frac{m(T_1 + T_2)}{2} = c$$

4.

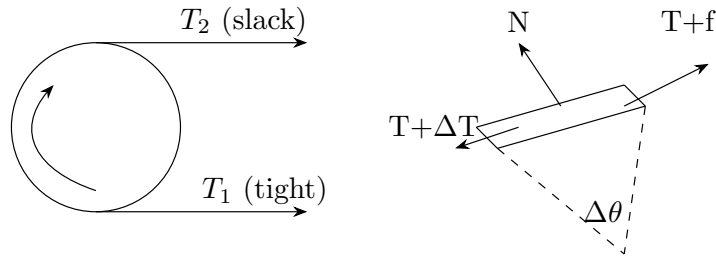
$$m_1(T_1 + T_2) = m_2(T_3 + T_4) = 2c$$

5.2 Belt Drives:

5.2.1 Contact Angle:



$$\cos\theta = \frac{R-r}{c} \implies \theta_{small} = 2\cos^{-1}\left(\frac{R-r}{c}\right), \theta_{big} = 2\pi - 2\cos^{-1}\left(\frac{R-r}{c}\right)$$



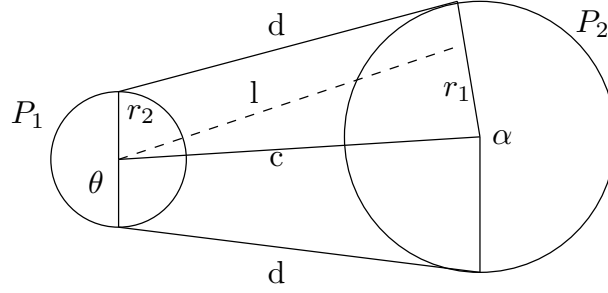
$$\Delta T = f = \mu N = \mu T \Delta\theta \implies \frac{\Delta T}{\Delta\theta} = \mu T$$

$$\frac{dT}{d\theta} = \mu T \implies \int_{T_2}^{T_1} \frac{dT}{T} = \mu \int_0^\theta d\theta \implies \boxed{\frac{T_1}{T_2} = e^{\mu\theta}}$$

5.2.2 Power Transfer:

$$P = \tau\omega = (T_1 r - T_2 r)\omega \implies \boxed{P = (T_1 - T_2)v}$$

5.2.3 Length of a Belt Drive:



$$d = \sqrt{c^2 + (r_1 - r_2)^2}$$

$$P_1 = r_2\theta, \quad P_2 = r_1\alpha = r_1(2\pi - \theta)$$

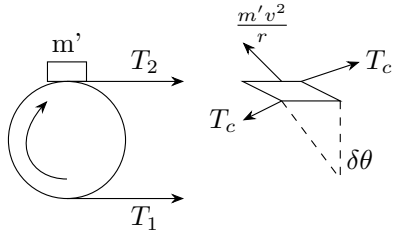
$$\therefore L = 2d + P_1 + P_2 = 2\sqrt{c^2 + (r_1 - r_2)^2} + r_2\theta + r_1(2\pi - \theta)$$

If c is very large,

$$\sqrt{c^2 + (r_1 - r_2)^2} = c\sqrt{1 - \left(\frac{r_1 - r_2}{c}\right)^2} = c\left(1 - \frac{(r_1 - r_2)^2}{2c^2}\right)$$

$$\Rightarrow \boxed{L = 2c - \frac{(r_1 - r_2)^2}{c} + r_2\theta + r_1(2\pi - \theta)}$$

5.2.4 Centrifugal Force:



If m is the mass per unit length,

$$T_c\delta\theta = \frac{m'v^2}{r} = \frac{mr\delta\theta}{r}$$

Hence,

$$\boxed{T_c = mv^2}$$

Actual Tensions:

$$1. \text{ Tight Side: } T_t = T_1 + T_c$$

$$2. \text{ Slack Side: } T_s = T_2 + T_c$$

$$\text{Let } e^{\mu\theta} = k \Rightarrow \frac{T_1}{T_2} = k \Rightarrow T_2 = \frac{T_1}{k}$$

$$\therefore \text{ Power: } P = (T_1 - T_2)v = T_1v\left(1 - \frac{1}{k}\right) = (T_t - mv^2)v\left(1 - \frac{1}{k}\right)$$

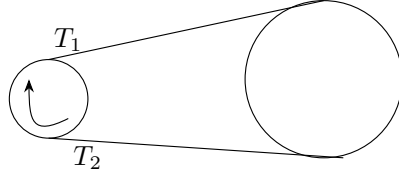
For maximum power, $P' = 0$

$$\frac{dP}{dv} = 0 \implies T_t = 3mv^2 \implies \boxed{T_t = 3T_c}$$

Maximum Tension:

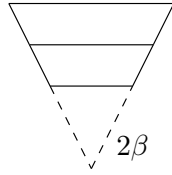
$$\boxed{(T_t)_{max} = \sigma_a wt}$$

where σ_a is allowable stress applied, w is width and t is thickness of the belt.



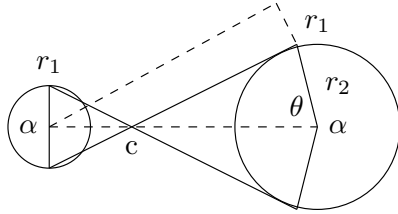
$$\text{Initial Tension: } T_o = \frac{T_1 + T_2}{2}$$

$\frac{T_1}{T_2} = e^{\mu\theta}$ is a limiting condition and is only possible at the verge of sliding.



$$\mu' = \frac{\mu}{\sin\beta} \implies \frac{T_1}{T_2} = e^{\mu'\theta} = e^{\frac{\mu}{\sin\beta}\theta}$$

5.2.5 Cross Belt Drives:



All equations are the same for cross belt drives as open drives except for the contact angle and length of drive equations.

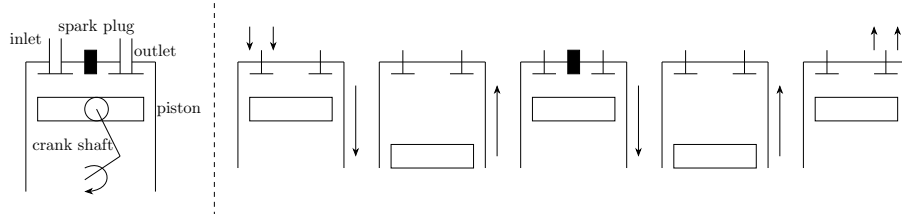
$$\text{By geometry, } \cos\theta = \frac{r_1 + r_2}{c}$$

$$\text{Contact Angles, } \alpha = 2\pi - 2\theta = 2\pi - 2\cos^{-1}\left(\frac{r_1 + r_2}{c}\right)$$

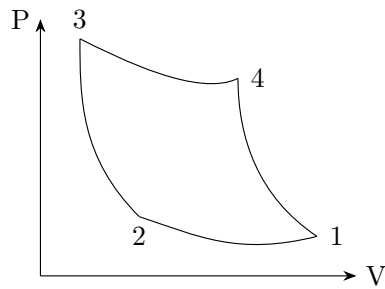
$$\text{Length of Belt, } L = 2\sqrt{c^2 - (r_1 + r_2)^2} + (r_1 + r_2)\alpha$$

6 Energy Conversion:

6.1 IC Engines:



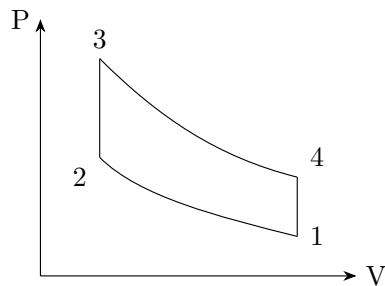
6.2 Carnot Cycle:



It is a reversible cycle.
Processes 12 and 34 are isothermal.
Processes 23 and 41 are isentropic.

$$\eta = 1 - \frac{T_{source}}{T_{sink}}$$

6.3 Otto Cycle:



Processes 12 and 34 are isentropic.
Processes 23 and 41 are isochoric.
Compression Stroke: Process 12.
Power Stroke: Process 34.

1. **Top Dead Center (TDC):** Topmost point of the piston in the engine.
2. **Bottom Dead Center (BDC):** Bottommost point of the piston.
3. **Bore Diameter (D):** Diameter of the cylinder.

4. **Stroke Length (L):** |TDC-BDC|.
5. **Displacement Volume (V):** $V = \frac{\pi}{4} D^2 L$.
6. **Mean Effective Pressure (MEP):**

$$P_m = \frac{\text{Net work Done}}{\text{Displacement Volume}}$$

6.3.1 Efficiency of a Otto Cycle:

If m is the mass of working fuel.

$$q_{in} = mC_v(T_3 - T_2) \quad q_{out} = mC_v(T_4 - T_1)$$

$$\eta_{otto} = 1 - \frac{q_{out}}{q_{in}} = 1 - \frac{T_4 - T_1}{T_3 - T_2}$$

For processes 12 and 34, $PV^k = c \implies TV^{k-1} = c$

$$\frac{T_3}{T_4} = \left(\frac{V_4}{V_3}\right)^{k-1} \quad \frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{k-1} = \left(\frac{V_4}{V_3}\right)^{k-1}$$

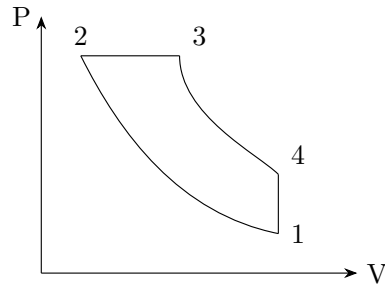
$$\therefore \eta_{otto} = 1 - \frac{T_1(T_4/T_1 - 1)}{T_2(T_3/T_2 - 1)} = 1 - \frac{T_1}{T_2} = \left(\frac{V_4}{V_3}\right)^{k-1}$$

If r is the compressibility ratio, then

$$\boxed{\eta_{otto} = 1 - \left(\frac{1}{r}\right)^{k-1}}$$

6.4 Diesel Cycle:

It doesn't have a spark plug; instead it has a fuel injection.



Processes 12 and 34 are isentropic.
Process 23 is isobaric.
Process 41 is isochoric.

$$q_{in} = mC_p(T_3 - T_2) \quad q_{out} = mC_v(T_4 - T_1)$$

$$\eta_{diesel} = 1 - \frac{q_{out}}{q_{in}} = 1 - \frac{C_v(T_4 - T_1)}{C_p(T_3 - T_2)}$$

$$\text{If } \frac{V_{max}}{V_{min}} = \frac{V_1}{V_2} = r, \quad \frac{V_3}{V_2} = r_c$$

where r is the compression ratio, and r_c is the cutoff ratio.

$$\eta_{diesel} = 1 - \frac{1}{k} \left(\frac{T_1}{T_2} \right) \left(\frac{T_4/T_1 - 1}{T_3/T_2 - 1} \right)$$

$$\therefore \boxed{\eta_{diesel} = 1 - \frac{1}{r^{k-1}} \left(\frac{r_c^k - 1}{k(r_c - 1)} \right)}$$

If $r_c \rightarrow 1$, then it becomes an Otto Cycle.

Question: An ideal diesel engine has a compression ratio of 20 and air as working fluid. The state of fluid at the beginning of compression process is 95 kPa and 20°C. If the maximum temperature of the cycle does not exceed 2200K, then determine the maximum pressure and thermal efficiency provided $k = 1.4$.

Solution: Given, $r = 20$, $P_1 = 95 \text{ kPa}$, $T_1 = 293 \text{ K}$, $T_3 = 2200 \text{ K}$

For process 12, $PV^k = c \implies TV^{k-1} = c$

$$\therefore \frac{T_2}{T_1} = \left(\frac{V_1}{V_2} \right)^{k-1} \implies T_2 = r^{k-1} T_1 = (20)^{0.4} (293) = 971.135 \text{ K}$$

$$\frac{P_2}{P_1} = \left(\frac{V_1}{V_2} \right)^k \implies P_2 = r^k P_1 = (20)^{1.4} (95) = 6297.46 \text{ kPa}$$

For process 23, $P = c \implies VT^{-1} = c$

$$\frac{T_3}{T_2} = \frac{V_3}{V_2} \implies r_c = \frac{2200}{971.135} = 2.265$$

$$\therefore \eta = 1 - \frac{1}{r^{k-1}} \left(\frac{r_c^k - 1}{k(r_c - 1)} \right) = 0.636 = 63.6\%$$